

Utilization of environmental and epidemiological indicators in the study of malaria dynamics

Raphael Felberg Levy^a, Flavio Codeço Coelho^a

^aSchool of Applied Mathematics, EMAP FGV, Praia de Botafogo 190, Rio de Janeiro, 22250-145, RJ, Brazil

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ABSTRACT

This paper aims to analyze the behavior of malaria transmission in the Amazon region based on climatic and environmental changes, such as temperature, precipitation, deforestation and forest fires, through proposed modifications to the SIR and SEI models, in order to contribute to the study of applications of external effects on the evolution of the disease. The Trajetorias Project, developed by the Synthesis Center on Biodiversity and Ecosystem Services (SinBiose/CNPq) was used as an initial reference for the study.

This work employs a modified SEIRS/SEI methodology, based on the SIR/SEI model developed by Parham and Michael (2010) which takes into account rainfall and temperature, with further modifications.

1. Introduction

Malaria remains one of the most significant vector-borne diseases worldwide, caused by protozoa of the genus *Plasmodium* and transmitted by female mosquitoes of the genus *Anopheles*. Although present in multiple regions, the burden of malaria in the Americas is highly concentrated in the Amazon basin, which spans nine countries and accounts for a substantial proportion of reported cases [1]. The persistence of malaria in this region is closely linked to ecological, climatic, and socioeconomic factors that influence both vector dynamics and human exposure.

Environmental changes play a central role in shaping malaria transmission. In the Amazon, processes such as deforestation, agricultural expansion, and infrastructure development alter local ecosystems, affecting vector habitats and increasing human–vector contact. These changes often promote the proliferation of mosquito breeding sites and concentrate human populations in high-risk areas, particularly among non-immune migrants [4, 5]. In addition to land-use change, climatic variables such as temperature and rainfall, as well as disturbances such as forest fires and mining activities, further influence vector abundance, pathogen development, and transmission intensity.

Despite the recognized importance of environmental drivers, integrating these factors into mathematical models of malaria transmission remains challenging. In particular, distinguishing between their effects on mosquito biology and their impact on human–vector interaction is essential for developing models that are both biologically consistent and identifiable from available data.

In this work, we develop and analyze a modified SIR–SEI

model for malaria transmission in the Amazon, incorporating environmental variables such as temperature, rainfall, deforestation, and fire activity. Climatic factors are modeled as drivers of mosquito life-history parameters, while land-use variables are introduced as modifiers of transmission intensity. Using epidemiological and environmental data from the Trajetorias Project [6], we investigate how these factors influence disease dynamics and assess their relative contributions through analytical and numerical methods.

2. Model Formulation

2.1. Baseline Delayed SIR–SEI Model

We begin with the classical host-vector framework for malaria transmission, based on the model introduced by Parham and Michael (2010) [7], which incorporates temperature- and rainfall-dependent mosquito dynamics. The human population is divided into susceptible, infected, and recovered compartments, denoted by S_H , I_H , and R_H , respectively. The mosquito population is structured into susceptible, exposed, and infectious compartments, denoted by S_M , E_M , and I_M .

The transmission dynamics are governed by the system of delay differential equations:

$$\frac{dS_H}{dt} = -ab_2 \frac{I_M(t)}{N} S_H(t) \quad (1a)$$

$$\frac{dI_H}{dt} = ab_2 \frac{I_M(t - \tau_H)}{N} S_H(t - \tau_H) - \gamma I_H(t) \quad (1b)$$

$$\frac{dR_H}{dt} = \gamma I_H(t) \quad (1c)$$

$$\frac{dS_M}{dt} = b(R, T) - \left(ab_1 \frac{I_H(t)}{N} + \mu(T) \right) S_M(t) \quad (1d)$$

$$\begin{aligned} \frac{dE_M}{dt} &= ab_1 \frac{I_H(t)}{N} S_M(t) - \mu(T) E_M(t) \\ &\quad - ab_1 \frac{I_H(t - \tau_M)}{N} S_M(t - \tau_M) \ell(\tau_M) \end{aligned} \quad (1e)$$

 raphael.levy147@gmail.com (R.F. Levy);

flavio.codeco.coelho@fgv.br (F.C. Coelho)

ORCID(s):

$$\frac{dI_M}{dt} = ab_1 \frac{I_H(t - \tau_M)}{N} S_M(t - \tau_M) \ell(\tau_M) - \mu(T) I_M(t) \quad (1f)$$

Here, $a(T)$ denotes the biting rate, b_1 and b_2 are transmission probabilities from humans to mosquitoes and vice versa, $\mu(T)$ is the mosquito mortality rate, and τ_H , τ_M represent the intrinsic and extrinsic incubation periods. The function $b(R, T)$ describes mosquito recruitment, while $\ell(\tau_M)$ represents survival during the extrinsic incubation period.

Although this formulation incorporates biologically relevant delays, numerical simulations revealed that, under the parameterization considered, the model fails to reproduce sustained transmission dynamics. In particular, both the infected human population and the mosquito population tends to extinction, indicating that the structure does not adequately capture endemic persistence.

2.2. ODE approximation and structural limitations

To simplify the analysis and avoid the technical complexity of delay differential equations, an ordinary differential equation (ODE) approximation was initially considered. In this formulation, the extrinsic incubation period in mosquitoes is modeled via a transition rate b_3 , replacing the explicit delay τ_M .

The resulting system is given by:

$$\frac{dS_H}{dt} = -ab_2 \frac{I_M}{N} S_H \quad (2a)$$

$$\frac{dI_H}{dt} = ab_2 \frac{I_M}{N} S_H - \gamma I_H \quad (2b)$$

$$\frac{dR_H}{dt} = \gamma I_H \quad (2c)$$

$$\frac{dS_M}{dt} = b(R, T) - \left(ab_1 \frac{I_H}{N} + \mu \right) S_M \quad (2d)$$

$$\frac{dE_M}{dt} = ab_1 \frac{I_H}{N} S_M - (\mu + b_3 \ell) E_M \quad (2e)$$

$$\frac{dI_M}{dt} = b_3 \ell E_M - \mu I_M \quad (2f)$$

While this formulation restores positivity and simplifies numerical implementation, it exhibits a fundamental limitation: the system tends toward a disease-free equilibrium after a single outbreak, failing to capture the recurrent nature of malaria transmission observed in endemic regions. This behavior highlights the importance of explicitly modeling incubation periods and immunity dynamics.

2.3. SEIRS-SEI model with ODE approximation

To address the limitations of the previous formulations, we introduce an ODE based SEIRS-SEI model that incorporates both incubation delays as done above and temporary immunity in the human population. The human

compartments are extended to include an exposed class E_H , and recovered individuals lose immunity at rate ω , returning to the susceptible class.

At this step, we included b_{3H} as the intrinsic incubation period $\left(b_{3H} = \frac{1}{\tau_H} \right)$, and substituted the original $b_3 = \frac{1}{\tau_M}$ for a Briere function $b_{3M} = \eta T(T - T_{\min})(T_{\max} - T)^{1/2}$, to take into consideration the minimum and optimal temperatures for development of the parasite [39, 40].

The scaling parameter η was estimated from the observed optimal sporogonic development rate through

$$\eta := \frac{1}{T_{\text{opt}}(T_{\text{opt}} - T_{\min})(T_{\max} - T_{\text{opt}})^{1/2}} \quad (3)$$

Another addition was the inclusion of a humidity-dependent survival component $p(H)$ on to the daily survival rate of the adult mosquitoes, which is modeled as a sigmoid function of the daily minimum relative humidity U . The parameter H_0 (in %) represents the humidity threshold at which the mosquito experiences a 50% reduction in survival relative to its maximum, and k (in $\%^{-1}$) controls the steepness of the transition. The parameter φ (dimensionless) establishes a baseline survival floor, accounting for the persistence of individuals in favorable micro-habitats even under extreme desiccation conditions.

The SEIRS-SEI complete model is then given by:

$$\frac{dS_H}{dt} = -\lambda_H(t) S_H + \omega R_H \quad (4a)$$

$$\frac{dE_H}{dt} = \lambda_H(t) S_H - b_{3H} E_H \quad (4b)$$

$$\frac{dI_H}{dt} = b_{3H} E_H - \gamma I_H \quad (4c)$$

$$\frac{dR_H}{dt} = \gamma I_H - \omega R_H \quad (4d)$$

$$\frac{dS_M}{dt} = \Lambda(t) - \lambda_M(t) S_M - \mu S_M \quad (4e)$$

$$\frac{dE_M}{dt} = \lambda_M(t) S_M - (\mu + b_{3M} \ell) E_M \quad (4f)$$

$$\frac{dI_M}{dt} = b_{3M} \ell E_M - \mu I_M \quad (4g)$$

The forces of infection are defined as:

$$\lambda_H(t) = a b_2 \frac{I_M(t)}{N(t)} \quad (5a)$$

$$\lambda_M(t) = a b_1 \frac{I_H(t)}{N(t)} \quad (5b)$$

Mosquito population dynamics were initially modeled using a linear birth–death process of the form

$$\frac{dM}{dt} = b - \mu M, \quad (6)$$

where b represents a recruitment rate, the birth rate in this case, and μ is the per capita mortality rate. However, this formulation does not account for environmental limitations on population growth.

To incorporate the effect of limited resources, we introduce a variant of density-dependent regulation inspired by the classical Verhulst (logistic) equation [37],

$$\frac{dM}{dt} = bM \left(1 - \frac{M}{K}\right) - \mu M, \quad (7)$$

where b denotes the per capita birth rate, μ the mortality rate, and K the environmental carrying capacity. This formulation separates recruitment and mortality mechanisms while capturing the reduction in population growth as the population approaches the environmental limit.

In the present model, this idea is adapted to mosquito recruitment by replacing the constant birth term with a density-dependent function,

$$\Lambda(t) = b(R, T) K(t) \left(1 - \frac{M(t)}{K(t)}\right) \quad (8)$$

where $M(t) = S_M + E_M + I_M$ and $K(t)$ is the environmental carrying capacity defined by

$$K(t) = M_0 \cdot f_T(T(t)) \cdot f_R(R(t)) \quad (9)$$

where M_0 is the initial mosquito population size (we assumed $M_0 = 35 \times N_0$ in our simulations, N_0 being the initial human population), and the environmental response functions are given by

$$f_T(T) = \min\left(1.0, \frac{T(T - T_{\min A.})\sqrt{T_{\max A.} - T}}{T_{\text{op A.}}(T_{\text{op A.}} - T_{\min A.})\sqrt{T_{\max A.} - T_{\text{op A.}}}}\right) \quad (10a)$$

$$f_R(R) = \left(\frac{R}{R_L/2}\right) \exp\left(1 - \frac{R}{R_L/2}\right) \quad (10b)$$

Unlike the classical logistic formulation, where growth is proportional to the current population size, this representation assumes that recruitment is primarily determined by environmental conditions and breeding site availability.

$f_R(R)$ increases with rainfall, making more breeding sites available, but decreases after reaching the threshold, flooding out the available habitats.

The term $\left(1 - \frac{M(t)}{K(t)}\right)$ ensures that recruitment decreases as the population approaches the carrying capacity, providing a biologically realistic upper bound on mosquito abundance. The mathematical implications of this formulation are analyzed in Section 5.1.

3. Data Description and Processing

3.1. Epidemiological Data

This study focuses on malaria cases caused by *Plasmodium vivax*, which accounts for the majority of malaria infections in Brazil and is the predominant species in the Amazon region [11, 12, 13].

Epidemiological data were obtained from the SIVEP-Malaria system [18], which provides daily records of notified malaria cases for all Brazilian municipalities from 2003 to 2023. The analysis was restricted to the period 2017–2023, for which complete epidemiological and environmental data were available.

3.2. Population Estimates

The municipality of Manaus was selected due to its high incidence of *P. vivax* malaria, particularly in rural areas. However, detailed yearly estimates of rural population were not directly available from census data. To address this limitation, population estimates were derived using data from the Trajetorias Project [6], which provides incidence values aggregated over five-year periods.

Incidence is defined as

$$\text{Inc}(d, m, z, t_1, t_2) = \frac{\text{Cases}(d, m, z, t_1, t_2)}{\text{Pop}(m, z, (t_1 + t_2)/2) \cdot (t_2 - t_1)} \times 10^5, \quad (11)$$

where $\text{Cases}(d, m, z, t_1, t_2)$ denotes the number of reported cases for disease d in municipality m and zone z over the interval $[t_1, t_2]$, and $\text{Pop}(m, z, \cdot)$ represents the population at the midpoint of the interval.

This relationship was inverted to estimate rural population values from reported incidence and case counts. Using this approach, the rural population of Manaus was estimated to represent less than 1% of the total population, ranging between 0.3% and 0.6% of the total municipal population, while accounting for a disproportionately large share of malaria cases (approximately 30% to 40% over the study periods).

Yearly total population values for Manaus were obtained via interpolation based on official census data from the Brazilian Institute of Geography and Statistics (IBGE) for the years 2000, 2010, and 2022 [15, 16, 17]. Intermediate yearly values were constructed using linear interpolation, while values beyond 2022 were extrapolated based on observed trends.

3.3. Data Consistency Adjustments

During preprocessing, inconsistencies were identified in the Trajetorias dataset. In particular, incidence rates for the period 2015–2019 had been calculated using population estimates corresponding to 2004–2008, resulting in a temporal mismatch. These values were recalculated using population estimates consistent with the appropriate time interval.

Additionally, rural and urban population estimates for intermediate years had been derived using fixed proportions based on 2010 data. To better account for demographic changes, particularly urbanization, these estimates were revised. The rural population for 2006 was approximated as the mean between the reported values for 2000 and 2010, while the 2017 value was obtained through extrapolation based on the observed growth trend. These adjustments ensured temporal consistency in the population data used throughout the study.

4. Impacts of Environmental Factors

Environmental changes, particularly deforestation and forest fires, have been widely associated with the resurgence and spatial redistribution of malaria transmission in the Amazon region [20]. The removal of forest canopy alters local microclimatic conditions, increasing sunlight exposure and ambient temperature, which in turn favor the development of mosquito larvae and pupae. In particular, the primary malaria vector in the Amazon, *Anopheles darlingi*, is known to thrive in partially deforested environments, where sunlit water bodies promote larval growth and increase the availability of food sources such as algae [21, 22].

Beyond ecological effects, deforestation also influences human exposure to malaria. Activities such as logging, agriculture, and infrastructure development lead to increased human presence in recently altered environments, often in close proximity to mosquito breeding sites. This results in higher human–vector contact rates and facilitates transmission, especially in areas where both vector density and environmental suitability are elevated. Additionally, land-use changes such as agriculture may contribute to river sedimentation and the formation of artificial breeding sites, further enhancing mosquito proliferation.

Forest fragmentation has been shown to accelerate mosquito development cycles by increasing temperature and light availability, reducing the time from larval to adult stages and increasing vectorial capacity and survivorship [27, 28]. Similarly, forest fires, often used as a land-clearing mechanism, remove vegetation and modify habitat structure, contributing to long-term ecological changes that can influence mosquito populations and disease transmission dynamics [29, 30]. These processes may also increase the likelihood of subsequent fires by promoting understory drying and fuel accumulation [31].

In this context, deforestation and fire activity are incorporated into the model as external drivers that modify transmission dynamics. Rather than directly altering mosquito physiological parameters, these factors are interpreted as influencing human–vector contact rates and exposure, thereby affecting the effective transmission of the disease.

Deforestation data were obtained from the DETER system, which provides daily records of deforestation alerts by degradation type, reported in km² [34]. The data were aggregated to produce a complete daily time series of deforested area. Forest fire data were obtained from the Brazilian National Institute for Space Research (INPE), which provides daily records of detected fire hotspots [35]. Due to the absence of area-based measurements at this temporal resolution, the number of detected hotspots was used as a proxy for fire activity.

5. Mathematical Analysis of the Model

5.1. Mosquito population dynamics

In this section, we analyze the qualitative impact of the density-dependent recruitment term on mosquito population dynamics. We compare the proposed formulation with the classical Verhulst model and show that the modified structure prevents extinction under suitable environmental conditions.

Proposition 1 (Environmental persistence of the mosquito population). *Assume that $b(R(t), T(t)) \geq b_0 > 0$ and $K(t) \geq K_0 > 0$ for all $t \geq 0$. Then, the mosquito population $M(t)$ governed by*

$$\frac{dM}{dt} = b(R, T) K(t) \left(1 - \frac{M(t)}{K(t)} \right) - \mu M(t) \quad (12)$$

cannot converge to zero as $t \rightarrow \infty$. In particular, $M = 0$ is not an equilibrium and cannot be asymptotically stable.

Proof. We begin by recalling the classical logistic model with mortality:

$$\frac{dM}{dt} = bM \left(1 - \frac{M}{K} \right) - \mu M. \quad (13)$$

In this formulation, the right-hand side vanishes at $M = 0$, so that $M = 0$ is always an equilibrium. Depending on the parameters, this equilibrium may be stable, allowing extinction of the population.

In contrast, in the present model the recruitment term is given by

$$\Lambda(t) = b(R, T) K(t) \left(1 - \frac{M(t)}{K(t)} \right), \quad (14)$$

and the population dynamics satisfy

$$\frac{dM}{dt} = b(R, T) K(t) \left(1 - \frac{M(t)}{K(t)} \right) - \mu M(t). \quad (15)$$

Evaluating the right-hand side at $M = 0$, we obtain

$$\left. \frac{dM}{dt} \right|_{M=0} = b(R, T) K(t). \quad (16)$$

By assumption, $b(R(t), T(t)) \geq b_0 > 0$ and $K(t) \geq K_0 > 0$, hence

$$\left. \frac{dM}{dt} \right|_{M=0} \geq b_0 K_0 > 0. \quad (17)$$

Therefore, $M = 0$ is not an equilibrium of the system.

Moreover, for $M(t)$ sufficiently small, we have

$$\frac{dM}{dt} \geq b_0 K_0 - \mu M(t). \quad (18)$$

Choosing $\varepsilon > 0$ such that $\varepsilon < \frac{b_0 K_0}{\mu}$, it follows that

$$\frac{dM}{dt} > 0 \quad \text{whenever } 0 < M(t) < \varepsilon.$$

Thus, any trajectory entering a neighborhood of zero is driven away from it.

Consequently, $M(t)$ cannot converge to zero as $t \rightarrow \infty$, and the extinction state is not asymptotically stable. $\square$

5.2. Basic Reproduction Number ($\mathcal{R}_0$)

To derive the $\mathcal{R}_0$ of the model, we define X_s as the set of all disease-free states,

$$X_s = \{x \geq 0 | x_i = 0, i = 1, \dots, m\},$$

where $X = (x_1, \dots, x_n)^T$, such that $x_i \geq 0$ represents the number of individuals in each compartment, and each function is assumed to be continuously differentiable at least twice in each variable (C^2) [36].

Rearranging the equations so that the first m contain the infected individuals, let $\mathcal{F}_i(x)$ be the rate of appearance of new infections in compartment i , $\mathcal{V}_i^+(x)$ be the rate of individuals entering compartment i by other means, and $\mathcal{V}_i^-(x)$ be the rate of individuals leaving compartment i . The disease transmission model consists of non-negative initial conditions together with the following system of equations:

$$\dot{x} = f_i(x) = \mathcal{F}_i(x) - \mathcal{V}_i(x), i = 1, \dots, n,$$

where $\mathcal{V}_i(x) = \mathcal{V}_i^-(x) - \mathcal{V}_i^+(x)$. Here, $F = \left[\frac{\partial \mathcal{F}_i(x_0)}{\partial x_j} \right]$ and $V = \left[\frac{\partial \mathcal{V}_i(x_0)}{\partial x_j} \right]$, where x_0 is a Disease-Free Equilibrium (DFE), and $1 \leq i, j \leq m$, and $\mathcal{R}_0 = \rho(FV^{-1})$.

The formulation for $\mathcal{R}_0$ in both model, as well as the complete SIR/SEI system is given by:

SIR: In this case, $m = 1$, and the compartments are arranged as $[I_H, S_H, R_H]$. Since $\mathcal{R}_0$ is calculated with normalized values, the necessary equations are multiplied by N to remove the denominator. Specifically, for the SIR case, as R_H is not used in any of the equations, it can be expressed solely in terms of S and I . Therefore:

- $\mathcal{F}_i(x)$: rate of entrance of infected individuals in compartment i
- $\mathcal{V}_i(x)^-$: rate of leaving compartment i
- $\mathcal{V}_i(x)^+$: rate of entering compartment i

So we have

$$\mathcal{F} = [ab_2 I_M S_H]$$

$$\mathcal{V}^- = [\gamma I_H]$$

$$\mathcal{V}^+ = [0]$$

$$\mathcal{V}_i(x) = \mathcal{V}_i(x)^- - \mathcal{V}_i(x)^+$$

Therefore,

$$\mathcal{V} = [I_H \gamma]$$

Hence

$$F = \frac{\partial \mathcal{F}}{\partial I_M} = [ab_2 S_H]$$

$$V = \frac{\partial \mathcal{V}}{\partial I_H} = [\gamma]$$

At the equilibrium, $[S_H^*, I_H^*] = [1, 0]$, so

$$F = [ab_2], V = [\gamma] \text{ and } \mathcal{R}_0 = \left| \frac{ab_2}{\gamma} \right|.$$

SEI: In this case, $m = 2$, and the compartments are arranged as $[E_M, I_M, S_M]$. Again, the necessary equations are multiplied by N to remove the denominator. Therefore:

$$\mathcal{F} = \begin{bmatrix} ab_1 I_H S_M \\ 0 \end{bmatrix}$$

$$\mathcal{V}^- = \begin{bmatrix} E_M(\mu + b_3 + l) \\ \mu I_M \end{bmatrix}$$

$$\mathcal{V}^+ = \begin{bmatrix} 0 \\ b_3 E_M \end{bmatrix}$$

$$\mathcal{V}_i(x) = \mathcal{V}_i(x)^- - \mathcal{V}_i(x)^+$$

Thus,

$$\mathcal{V} = \begin{bmatrix} E_M(\mu + b_3 + l) \\ \mu I_M - b_3 E_M \end{bmatrix}$$

Hence

$$\begin{aligned} F &= \frac{\partial \mathcal{F}}{\partial E_M, I_H} \\ &= \begin{bmatrix} \frac{\partial ab_1 I_H S_M}{\partial E_M} & \frac{\partial ab_1 I_H S_M}{\partial I_H} \\ \frac{\partial 0}{\partial E_M} & \frac{\partial 0}{\partial I_H} \end{bmatrix} \\ &= \begin{bmatrix} 0 & ab_1 S_M \\ 0 & 0 \end{bmatrix} \end{aligned}$$

$$\begin{aligned}
V &= \frac{\partial \mathcal{V}}{\partial E_M, I_M} \\
&= \begin{bmatrix} \frac{\partial E_M(\mu + b_3 + l)}{\partial \mu I_M - b_3 E_M} & \frac{\partial E_M(\mu + b_3 + l)}{\partial I_M} \\ \frac{\partial E_M}{\partial \mu I_M - b_3 E_M} & \frac{\partial E_M}{\partial I_M} \end{bmatrix} \\
&= \begin{bmatrix} \mu + b_3 + l & 0 \\ -b_3 & \mu \end{bmatrix}
\end{aligned}$$

At the equilibrium, $[S_M^*, E_M^*, I_M^*] = [1, 0, 0]$, so

$$F = \begin{bmatrix} 0 & ab_1 \\ 0 & 0 \end{bmatrix},$$

$$V = \begin{bmatrix} \mu + b_3 + l & 0 \\ -b_3 & \mu \end{bmatrix}$$

and

$$\mathcal{R}_0 = \left| \frac{ab_1 b_3}{(b_3 + l + \mu)\mu} \right|.$$

SIR/SEI: In this case, $m = 3$, and the compartments are arranged as $[I_H, E_M, I_M, S_H, S_M]$. Once again, the necessary equations are multiplied by N to remove the denominator. Therefore:

$$F = \begin{bmatrix} ab_2 I_M S_H \\ ab_1 I_H S_M \\ 0 \end{bmatrix}$$

$$\mathcal{V}^- = \begin{bmatrix} \gamma I_H \\ E_M(\mu + b_3 + l) \\ \mu I_M \end{bmatrix}$$

$$\mathcal{V}^+ = \begin{bmatrix} 0 \\ 0 \\ b_3 E_M \end{bmatrix}$$

$$\mathcal{V}_i(x) = \mathcal{V}_i(x)^- - \mathcal{V}_i(x)^+$$

Thus,

$$\mathcal{V} = \begin{bmatrix} I_H \gamma \\ E_M(\mu + b_3 + l) \\ \mu I_M - b_3 E_M \end{bmatrix}$$

Hence

$$F = \frac{\partial F}{\partial I_H, E_M, I_M} = \begin{bmatrix} 0 & 0 & ab_2 S_H \\ ab_1 S_M & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$V = \frac{\partial \mathcal{V}}{\partial I_H, E_M, I_M} = \begin{bmatrix} \gamma & 0 & 0 \\ 0 & b_3 + l + \mu & 0 \\ 0 & -b_3 & \mu \end{bmatrix}$$

At the equilibrium,

$$[S_H^*, S_M^*, I_H^*, E_M^*, I_M^*] = [1, 1, 0, 0, 0],$$

so

$$F = \begin{bmatrix} 0 & 0 & ab_2 \\ ab_1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix},$$

$$V = \begin{bmatrix} \gamma & 0 & 0 \\ 0 & b_3 + l + \mu & 0 \\ 0 & -b_3 & \mu \end{bmatrix}$$

and

$$\mathcal{R}_0 = \left| \sqrt{\frac{a^2 b_1 b_2 b_3}{(b_3 + l + \mu)\gamma\mu}} \right| = \sqrt{\mathcal{R}_{0SIR} \times \mathcal{R}_{0SEI}}$$

5.3. Understanding relations

Having the necessary epidemiological and environmental datasets ready for use, the next step taken was to verify the correlation between the number of cases and forest areas impacted by deforestation and fires, in addition to the correlation with temperature and precipitation as well.

Using the Pandas library on a Python environment, the correlation between monthly number of cases caused by *P. vivax* in Manaus and deforested area was found to be approximately -0.11, and 0.137 between cases and burnt area. In relation to climatic data, the correlation with temperature was found to be around 0.404, and with precipitation to be -0.365. Both correlation tables were taken in relation to monthly data.

On the other hand, taking the correlation between the incidence of cases in relation to the seasons, deforested and burnt areas, that was found to be 0.722, 0.314 and 0.818, respectively. Looking for the correlation in relation to seasons, temperature and precipitation, that was found to be 0.687, 0.839 and -0.883, respectively.

6. Computational Implementation

All model simulations were performed using Python version 3.12.7 (conda-forge distribution). Data manipulation and analysis were performed using Pandas version 3.0.2, while numerical computations and array operations were carried out with NumPy version 2.4.2.

The systems of time-delay ordinary differential equations (1-6) were solved numerically using the explicit Runge-Kutta method as implemented in the `solve_ivp` function from

SciPy version 1.17.1. All computations were performed on a laptop running **Windows 11** (64-bit) equipped with a **12th Gen Intel Core i5-1235U** processor (10 cores, 12 threads, 1.30 GHz base frequency) and **16 GB RAM**.

The source code and data required to reproduce all figures are available at https://github.com/RaphaLevy/Study_on_malaria_dynamics_v2.

7. Results

The initial implementation of the model using the original functional forms and parameter values reported in [7] did not reproduce biologically consistent transmission dynamics, particularly in the mosquito population. As illustrated in Figures 1–3A, the human SIR component exhibits a monotonic decline in the number of infected individuals, suggesting that the infection fails to persist in the population under the given parametrization. More critically, the corresponding SEI mosquito model yields nonphysical behavior, with the susceptible mosquito population attaining negative values and exhibiting oscillatory dynamics.

These inconsistencies arise from the functional forms adopted for key entomological parameters, which do not explicitly enforce biologically meaningful bounds. In particular, probability and rate functions did not explicitly present necessary constraints, such as not ensuring that probabilities lie between $[0, 1]$, leading to violations of positivity and instability in the numerical solutions.

To address this issue, the functional forms were modified to ensure that all probabilities and rates remain within their admissible ranges. The resulting dynamics, shown in Figure 3A, exhibit a different behavior, with the mosquito population rapidly declining toward extinction. While this modification restores mathematical consistency, it also highlights a mismatch between the model structure and the expected persistence of malaria in endemic settings, motivating the need for further refinement of the transmission and demographic assumptions.

$$R_H(t) = \max(0, N(1 + r_H t) - S_H(t) - I_H(t)) \quad (20)$$

$I_H(t)$ now being the moving sum of cases over a period of 120 days, as it is the average duration of infectiousness, and $R_H(t)$ being the difference between the initial population and the susceptible and infected population at each day, with the addition of the daily births. The result can be seen in the graph below.

Before including the environmental factors on the used model (7-12), its results were compared to a model based on SIVEP data, keeping use of equations (7, 10-12), but using

$$I_H(t) = \sum_{\tau=\max\left(0, k-\frac{1}{\gamma}+1\right)}^{0 \leq k \leq 1825} \# \text{ of cases} \quad (19)$$

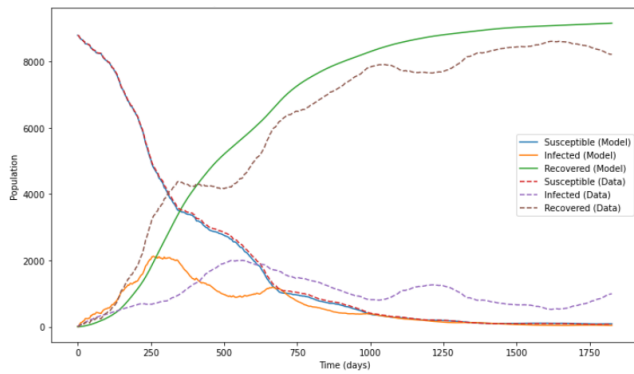


Figure 1: Comparison between observed data of infections and model estimation of I_H .

It is possible to see that the model gives a close approximation to the format of the observed number of infections and recoveries, although the infections appear to increase faster and die out given the model, a behavior that is not seen on the actual data.

At this point, a linear combination was implemented as a replacement to the use of simply T and R , in order to take into consideration the effects of deforestation and fires, with the α -function $\alpha_1 D + \alpha_2 F + \alpha_3 T + \alpha_4 R$, with $\alpha_0 = [0, 0, 1, 1]$ representing the primary version of the model.

8. Discussion

The population estimates provided by the Trajetórias Project [6] for inter-census periods are derived using fixed rural–urban proportions based on the 2010 census. While this approach ensures internal consistency, it does not account for the significant demographic changes observed in Manaus over time, particularly those associated with rapid urbanization.

To address this limitation, the population proportions used in this study were recalculated (see Section 3.3) using interpolation and extrapolation based on available census data. This adjustment provides a more realistic representation of temporal changes in the rural population, which is especially relevant given the disproportionate contribution of rural areas to malaria incidence.

These refinements improve the consistency between demographic inputs and epidemiological dynamics in the model, reducing potential biases in the estimation of transmission intensity and enhancing the reliability of the results.

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A. Appendix

A.1. Tables

Table A.1

List of model parameters.

Parameter	Definition	Formulation
$p(T)$	Daily mosquito survival rate dependent on temperature	$e^{(-1/(AT^2+BT+C))}$
$p(H)$	Daily mosquito survival rate dependent on humidity ¹	$\frac{\varphi e^{-k(H-H_0)+1}}{e^{-k(H-H_0)+1}+1}$
$p(T, H)$	Daily mosquito survival rate	$p(T)p(H)$
$p_L(R)$	Probability of larval survival dependent on rainfall ²	$(4p_{ML}/R_L^2)R(R_L - R)$
$p_L(T)$	Probability of larval survival dependent on temperature	$e^{-(c_1T+c_2)}$
$p_L(R, T)$	Probability of larval survival	$p_L(R)p_L(T)$
$\mu(T, H)$	Mosquito mortality rate per capita (days ⁻¹)	$-\log(p(T, H))$
$a(T)$	Biting rate (days ⁻¹)	$(T - T')/D_1$
$\tau_M(T)$	Duration of sporozoite cycle (days)	$DD/(T - T_{\min})$
$\tau_L(T)$	Duration of larval development phase (days)	$1/(c_1T + c_2)$
$\ell(\tau_M)(T, H)$	Probability of mosquito survival during sporozoite cycle (days ⁻¹)	$p(T, H)^{\tau_M(T)}$
$b(R, T)$	Mosquito birth rate (days ⁻¹)	$B_E p_E(R) p_L(R, T) p_P(R) / (\tau_E + \tau_L(T) + \tau_P)$
$M(t)$	Total population of mosquitos	$S_M(t) + E_M(t) + I_M(t)$
$N(t)$	Total population of humans	$S_H(t) + E_H(t) + I_H(t) + R_H(t)$

¹ $p(H)$ was added in order to take into consideration the deleterious effects of humidity. Unlike for T and R , which use the daily average value of temperature and precipitation, respectively, H uses the daily minimal value of humidity.

² $p_E(R)$ and $p_P(R)$ have an analogous expression to $p_L(R)$, substituting p_{ML} for p_{ME} and p_{MP} respectively.

Table A.2

List of model variables.

Symbol	Definition	Units
b_1	Proportion of bites from susceptible mosquitoes on infected humans that result in infection	Dimensionless
b_2	Proportion of bites from infected mosquitoes on susceptible humans that result in infection	Dimensionless
γ	1/Average duration of infectiousness in humans	days ⁻¹
B_E	Number of eggs laid per adult per oviposition	Dimensionless
p_{ME}	Maximum probability of egg survival	Dimensionless
p_{ML}	Maximum probability of larval survival	Dimensionless
p_{MP}	Maximum probability of pupal survival	Dimensionless
τ_E	Duration of the egg development phase	days
τ_P	Duration of the pupal development phase	days
τ_H	Latent period of infection within humans	days
R_L	Rainfall threshold until breeding sites are eliminated, removing immature individuals: 32.26 ³	mm
$T_{\min}$	Minimum temperature, below which there is no development of the parasite: 14.5	°C
$T_{\max}$	Maximum temperature, above which there is no development of the parasite: 29.8 [9]	°C
T_{opt}	Optimal temperature for development of the parasite: 23.5 [9]	°C
$T_{\min A.}$	Minimum temperature, below which there is no development of the vector: 16 [39, 41]	°C
$T_{\max A.}$	Maximum temperature, above which there is no development of the vector: 34 [39, 41, 42]	°C
$T_{\text{op } A.}$	Optimal temperature for transmission: 34 [39]	°C
DD	Degree-days for parasite development. "Sum of heat" for maturation: 105 [9, 10]	°C days
A	Empirical sensitivity parameter	(°C ² days) ⁻¹
B	Empirical sensitivity parameter	(°C days) ⁻¹
C	Empirical sensitivity parameter	days ⁻¹
D_1	Temperature scaling factor of the biting rate	°C days
c_1	Empirical sensitivity parameter	(°C days) ⁻¹
c_2	Empirical sensitivity parameter	days ⁻¹
T'	Biting rate temperature threshold ⁴	°C
b_{3M}	Infection rate in exposed mosquitoes ($1/\tau_M(T)$)	days ⁻¹
b_{3H}	Infection rate in exposed humans ($1/\tau_H$)	days ⁻¹
φ	Baseline survival probability at extreme low humidity	Dimensionless
H_0	Critical humidity threshold for 50% survival transition	%
k	Steepness of the humidity-survival sigmoid response	% ⁻¹

³ The reference rainfall level R_L for the Amazon was approximated based on reported associations between monthly precipitation and malaria incidence [8]. In particular, precipitation levels in the range of 917 to 1019 mm/month were associated with a decrease in malaria cases. The midpoint of this interval (968 mm/month) was used as a representative value and converted to a daily rate by dividing by 30 days, yielding $R_L \approx 32.3$ mm/day. This value is therefore interpreted as the threshold scale for high rainfall conditions in the model.

⁴ In Parham and Michael [7], this parameter is denoted T_1 . We rename it here to avoid confusion with the mean temperature in the absence of seasonality, also denoted T_1 in the original paper.

A.2. Figures

Malaria SIR Model with fixed probabilities and Parameters: $I_{H0} = 944$, $E_{M0} = 500$, $A = -0.03$, $B = 1.31$, $C = -4.4$, $\gamma = \frac{1}{120}$, $R_L = 50$, $T = 19.9$

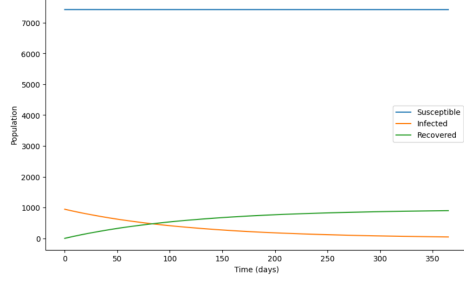


Figure 1A: Simulation of the baseline SIR model using the original parameterization from [7]. The dynamics show a monotonic decline in the number of infected individuals, indicating the absence of sustained transmission.

Malaria SEI Model with Parameters: $I_{H0} = 944$, $E_{M0} = 500$, $A = -0.03$, $B = 1.31$, $C = -4.4$, $\gamma = \frac{1}{120}$, $R_L = 50$, $T = 19.9$

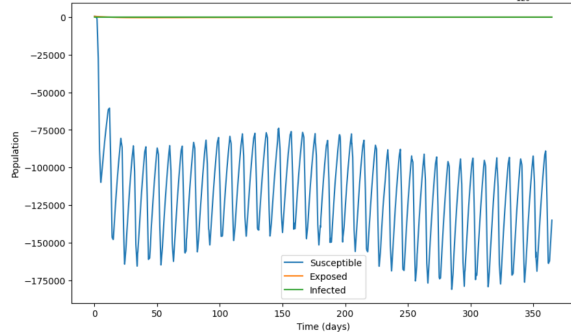


Figure 2A: Simulation of the baseline SEI mosquito model using the original parameterization from [7]. The susceptible mosquito population exhibits oscillatory behavior and attains negative values, indicating a loss of biological and mathematical consistency.

Malaria SEI Model with fixed probabilities and Parameters: $I_{H0} = 944$, $E_{M0} = 500$, $A = -0.03$, $B = 1.31$, $C = -4.4$, $\gamma = \frac{1}{120}$, $R_L = 50$, $T = 19.9$

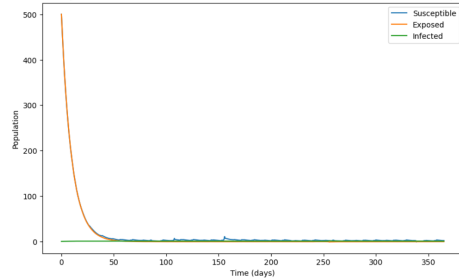


Figure 3A: Simulation of the SEI mosquito model after modifying parameter functions to enforce biologically admissible bounds on probabilities and rates. Although positivity is restored, the mosquito population rapidly declines toward extinction, suggesting that additional mechanisms are required to sustain endemic transmission.